INDIVIDUAL REFLECTION

Mini Use Case 02 – Credit Pre-Screening

Calibo AI Academy | Part 2 – KLU Batch | Phase 2 | Supervised Learning Part 1

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Role	Model Builder
Contribution	Step 5.4: Logistic Regression Pre-Screener and Step 5.5: Auto-Decline Threshold

Business objective

Use six months of historical loan decisions to identify clear-cut declines that may be removed from manual review, while limiting the risk of wrongly turning away creditworthy applicants. The aim is to balance safe screening with analyst time savings—not simply to maximize model accuracy.

Step 5.4 — Logistic Regression Pre-Screener

1. Target and features. The target was encoded as Declined = 1 and Approved = 0. Features included credit score, new-to-credit flag, FOIR, loan-to-income, delinquencies, active loans, job tenure, age, loan tenure, city tier, employment type and loan purpose. Decision was the target, not a feature; application ID, date, city and gender were excluded from model inputs.

2. Train-test split. An 80/20 stratified split with random_state=42 produced 9,600 training rows and 2,400 test rows. Training data fitted the model and preprocessing; the held-out test data measured performance.

3. Preprocessing. Missing income was filled using employment-type medians with an overall fallback. Missing credit scores were filled with the training median while the is_ntc flag preserved the fact that those applicants had no score. FOIR = (existing EMI + proposed EMI) / monthly income; loan-to-income = loan amount / (monthly income × 12). Numeric features were scaled and employment type and loan purpose were one-hot encoded. A pipeline kept these transformations together with Logistic Regression.

Model results at the default 0.50 threshold

Metric	Result	Meaning
Accuracy	79.08%	Share of all test predictions that were correct.
ROC-AUC	0.867	Ability to rank declines above approvals.
Precision — Declined	79.11%	Share of predicted declines that were actually declined historically.
Recall — Declined	75.47%	Share of actual declines identified by the model.

Confusion matrix

Actual	Predicted Approved	Predicted Declined
Approved	1,046 (TN)	225 (FP)
Declined	277 (FN)	852 (TP)

The key risk at the default threshold is the 225 false positives: applicants historically approved by the bank but predicted as declined. This is why accuracy alone is insufficient for a lending decision.

Important coefficients

Feature	Coefficient	Interpretation
FOIR	+1.22	Higher repayment obligations relative to income push toward decline.
Credit score	–1.01	Higher scores push toward approval.
Delinquencies in 24 months	+0.98	More missed or late repayments push toward decline.
Gig / Contract employment	+0.62	This category pushes toward decline in the fitted model.
Salaried employment	–0.45	This category pushes toward approval in the fitted model.

These coefficients describe patterns in historical decisions; they do not prove that a feature causes a decline.

Step 5.5 — Auto-Decline Threshold

The model produces a probability of decline. A threshold determines how high that probability must be before an application is flagged for automatic decline. The assignment requires that no more than 5% of the auto-declined group were historically approved.

Threshold	Auto-declined	Wrongly declined
0.50	44.88%	20.89%
0.60	37.62%	16.83%
0.70	30.75%	12.06%
0.80	23.46%	7.10%
0.85	19.67%	5.93%
0.90	15.21%	4.66%
0.95	10.00%	2.08%

Threshold 0.90 is selected because it is the lowest tested threshold that meets the 5% rule. At 0.85, the wrong-decline rate is 5.93%, so it fails the rule. A threshold of 0.95 is safer on this test set but screens out fewer applications.

Monthly business impact

Applications screened: 3,000 × 15.21% ≈ 456 applications per month.

Analyst hours saved: 456 × 25 minutes ÷ 60 ≈ 190 hours per month.

Observed wrong-decline rate: 4.66% (17 of 365 auto-declined test applications were historically approved). Applying this percentage to future applications is an estimate, not a guarantee.

> **New-to-credit safeguard:** The test results include 12 new-to-credit applicants in the automatic-decline group. Since accuracy for this group is only 56.7%, they should remain with human analysts. If this safeguard is applied, the screening rate and monthly savings must be recalculated.

Final summary

My contribution covered Steps 5.4 and 5.5. I built and evaluated a Logistic Regression classifier to predict historical loan decisions, using a stratified 80/20 split and a preprocessing pipeline for missing values, feature engineering, encoding and scaling. At the default 0.50 threshold, the model achieved 79.08% accuracy and 0.867 ROC-AUC, but 225 historically approved applicants were incorrectly predicted as declines. I then compared seven thresholds and selected 0.90, the lowest tested threshold meeting the bank's 5% wrong-decline limit: 4.66% of the auto-declined test group had been historically approved. At the stated business volume, this corresponds to about 456 applications and 190 analyst hours per month. Before deployment, new-to-credit applicants should be routed to human review, the resulting savings should be recalculated, and the model should be validated on fresh data and actual repayment outcomes.